

CAMBIUM ACADEMY · COURSE 03

Research with AI

Find real sources. Catch the fake ones.

July 2026 edition. Free forever, no signup. Certificate included when you pass.

Built for everyone who looks things up

- ◆ **No code, no jargon, no library science degree**

If you have ever asked AI a question and wondered whether to believe it, you are ready.

- ◆ **A browser, 90 minutes, and \$0**

That is the whole entry fee.

- ◆ **Works with any AI you already use**

ChatGPT, Claude, Gemini, Perplexity, or whatever is open in the next tab.

- ◆ **Made for a very specific moment**

The moment an AI answer looks perfect and you have no idea if it is true.

OUTCOMES

Six skills you will walk away with

◆ **Tell a grounded answer from a guess**

Some AI answers come with receipts. Most do not. You will know which is which.

◆ **Verify any claim in about 60 seconds**

Lateral reading, the same move professional fact-checkers use.

◆ **Catch a fabricated citation**

The fakes are engineered to look real. You will open them anyway, and know why.

◆ **Use deep research tools without being fooled**

A 20-source report is a first draft with leads, not a finished answer.

◆ **Cite honestly and disclose properly**

When AI helped, you will know what to check and what to say.

◆ **Stay the smartest thing in the room**

Judgment is the one research skill that keeps its value as models improve

YOUR PATH

Five steps, one certificate

1 Read the lecture slides

6 modules, about 25 min

2 Clear the 24 flashcards

Quick recall drills

3 Play the 3 AI Lab games

Catch fakes with your own eyes

4 Pass the quiz

20 questions, 14 to pass

5 Print your certificate

Instantly, ready for LinkedIn

One tip: open your favorite AI in another tab now. This is a doing course, and the whole point is to try these checks on a real answer.

MODULE 1 OF 6

The trust problem

The same AI that finds your sources is the biggest new source of fake ones.

Fabricated references, measured at scale

1 in 2,828

papers contained a fabricated reference in 2023

1 in 277

papers by early 2026, a roughly 6x rise in three years

2M+

papers scanned in the 2026 Lancet analysis

97M

citations checked to get these numbers

Dated fact, not a vibe: Lancet, 2026 (reported by STAT News, May 7, 2026). The rise tracks exactly the years AI writing assistants went mainstream.

The best research tool ever built has one flaw

- ◆ **It finds sources in seconds**

Real papers, real articles, summarized instantly. That part is genuinely revolutionary.

- ◆ **It also invents sources in seconds**

Same tool, same confident tone, and the fakes are formatted exactly like the finds.

- ◆ **You cannot tell them apart by looking**

A fabricated reference has a plausible author, a real-sounding journal, and a well-formed DOI.

Speed is not the problem. Trust is. This course is about knowing which answers deserve it.

An assistant who never says "I could not find it"

"AI is a brilliant, tireless research assistant with one dangerous habit: it would rather hand you something that looks right than admit it found nothing."

- ◆ **It is not lying, exactly**

It completes patterns. When the true answer is missing, a plausible-looking answer fills the gap.

- ◆ **The gap-filling looks like research**

Fluent, cited, formatted, confident. That is what makes it dangerous.

- ◆ **Your job is not to distrust everything**

It is to know which claims are load-bearing, and to check those.

Hold onto this picture. Every technique in the next five modules follows from it.

What this course trains

◆ Find

Get answers with receipts: search-grounded tools that retrieve first and cite as they go.

◆ Verify

Lateral reading and the SIFT habit: check claims across the web, fast.

◆ Cite

Open every citation that matters. Catch the fabricated ones before they catch you.

◆ Judge

Deep research tools, disclosure, and when checking is worth the time.

By the end, "the AI said so" will never again be your final answer on anything that matters.

MODULE 2 OF 6

Find: receipts, not vibes

Two kinds of AI answers. Only one shows where it got things.

Memory answers vs receipt answers

- ◆ **A plain chatbot answers from memory**

It compresses what it read during training and reconstructs an answer. No live lookup, no links.

- ◆ **A search-grounded tool retrieves, then answers**

It searches the web right now, reads real pages, and cites them as it writes.

- ◆ **Same question, different trust level**

The memory answer might be right. The grounded answer shows you where to check.

This is the single most useful thing to know about AI research in 2026: ask a tool that shows its receipts.

Tools that show their receipts

Perplexity

Built around search. Every claim in the answer carries a numbered, clickable source.

ChatGPT with search

Toggles into retrieval mode and links as it writes. The plain mode still answers from memory.

Gemini Deep Research

Browses dozens of pages and returns a long, cited report you can audit.

Names and buttons change monthly. The test never does: can you click through to where each claim came from?

Ask for the receipts, out loud

“Answer with sources I can click, and if you cannot find a real source for a claim, say so instead of guessing.”

- ◆ **One sentence changes the behavior**

You are telling it that receipts matter more than fluency.

- ◆ **Grounded tools do this by default**

Plain chatbots will at least flag what they are unsure of.

- ◆ **It is not a magic spell**

It is a standing expectation, the same one you would set with a new assistant.

A link is not proof by itself. Module 3 is about what to do after you get the receipts.

The two-tab test

◆ Tab 1: ask a plain chatbot

"What did the 2026 Lancet analysis find about fabricated references?" Watch it answer from memory.

◆ Tab 2: ask a grounded tool the same thing

Watch it search, then answer with numbered links.

◆ Compare the receipts

Which answer could you hand to a skeptical boss? Which one could you even begin to check?

Sixty seconds, and you will never unsee the difference. This is the AI Lab's first game, too.

MODULE 3 OF 6

Verify: lateral reading

Stop reading down the page. Start reading across the web.

Confident, cited, and wrong

- ◆ **Columbia Journalism Review ran the test**

Eight AI search tools, asked to identify the source of real news excerpts.

- ◆ **More than 60% of answers were wrong**

Wrong article, wrong outlet, or a link that did not support the claim.

- ◆ **The tone never wobbled**

The wrong answers sounded exactly as confident as the right ones.

The lesson is not "never use AI search." It is that confidence and citations are style, not evidence. Verification is a separate step, and it is yours.

Read across, not down

◆ **Reading down: the amateur habit**

Stay on one page, judge it by its own design, tone, and About page. Slow, and easy to fool.

◆ **Reading across: the fact-checker habit**

Leave the page. Open a new tab. See what independent sources say about the claim and the source.

◆ **It is faster, not slower**

Professional fact-checkers verify in minutes because they never trust a page to grade itself.

This is called lateral reading, and it is the single highest-value research habit of the decade.

SIFT: four moves, in order

S Stop

Before you share, cite, or believe: pause. Do you actually know this source and this claim?

I Investigate the source

New tab. Who is behind this? What do others say about them? Thirty seconds, not thirty minutes.

F Find better coverage

Search the claim itself. Do independent, reliable sources report the same thing?

T Trace to the original

Follow quotes, stats, and studies upstream to their first appearance. Context changes everything.

SIFT comes from digital literacy researcher Mike Caulfield. It was built for the open web, and it works even better on AI answers.

One claim, one new tab, one independent source

“Take the one claim your decision rests on. Open a new tab. Search it. Find one independent source that confirms it, before you build on it.”

- ◆ **Not every claim, just the load-bearing one**

The stat in your report, the quote in your post, the number in your decision.

- ◆ **Independent means independent**

A second AI summarizing the same guess does not count. Find the original, or reliable coverage of it.

- ◆ **Usually takes less than a minute**

One opened tab puts you ahead of almost everyone.

You will practice exactly this in the AI Lab's lateral reading race, against the clock.

Confidence is not accuracy

- ◆ **Wrong answers sound like right ones**

Polished, structured, sure of itself. The tone never tells you anything.

- ◆ **So decide by cost, not confidence**

Wrong movie trivia? Enjoy. Wrong stat in your thesis, your report, your medical question? Verify.

- ◆ **Make it a reflex**

High stakes means lateral check. No exceptions, no matter how good the answer looks.

Rule of thumb: the more you want the answer to be true, and the more it costs to be wrong, the more it deserves the 60-second check.

MODULE 4 OF 6

The citation trap

The fakes are engineered to pass. Here is how to catch every one.

How often AI invents citations

◆ Literature-review tasks, 2026 benchmarks

Frontier models still fabricate 18 to 28% of citations on these tasks. Roughly one in four.

◆ Even the best models

Sit at about 3 to 19% depending on the task. Better, and nowhere near zero.

◆ The trend is real but slow

Each generation fabricates less. None of them, so far, reliably says "I could not find a source."

Read that again: in 2026, a careful, state-of-the-art model can still hand you a reference list where one entry in five does not exist.

Citation-shaped text

- ◆ **The model learned what citations look like**

Author names, a year, a journal, a DOI. A pattern, learned from millions of real examples.

- ◆ **When unsure, it completes the pattern**

It does not look up a reference. It generates something citation-shaped that fits the topic.

- ◆ **The result is a convincing forgery**

Real researcher names on papers they never wrote. Real journals with invented articles. DOIs that resolve to nothing, or to something else.

Nobody programmed it to deceive. It is doing exactly what it was trained to do: produce plausible text. Plausible is the problem.

Anatomy of a fabricated citation

Plausible authors

Often real researchers from the right field, who never wrote this particular paper.

Real-sounding venue

An actual journal, or a perfect pastiche of one. Nothing about the name looks off.

A well-formed DOI

Formatted perfectly. It either resolves to a different paper or to nothing at all.

The right year and pages

Every field filled in, every convention followed. It is built to survive a glance.

A fabricated citation is designed, by the statistics of its training, to pass the eyeball test. So the eyeball test is worthless.

The only rule in this course written in stone

“A citation you never opened is a citation you cannot trust.”

- ◆ **Click every reference that matters**

Does the page exist? Is it the claimed authors, venue, and year?

- ◆ **Then check what it says**

Existing is not enough. Does the source actually support the claim it was attached to?

- ◆ **No exceptions for confident tone**

The fakes and the finds are formatted identically. Opening them is the only test that works.

This one habit, applied to just the references you actually rely on, catches essentially every fabricated citation.

Thirty seconds per citation

◆ Click 1: does it exist?

Paste the title into Google Scholar or the DOI into your browser. No result is your answer.

◆ Click 2: is it what it claims?

Right authors, right journal, right year? Fakes often borrow a real title with the wrong details.

◆ Click 3: does it say what the AI said?

Open the abstract. Does this source actually support the claim it was cited for?

Scale it sensibly: for a quick errand, check the one source that matters. For anything you will publish, submit, or decide on, check them all.

The niche-topic test

◆ Ask any chatbot for three sources

Pick a genuinely niche topic you know well. Narrow beats famous: fakes cluster where training data is thin.

◆ Open all three

Not skim. Open. Scholar for papers, the DOI for articles, the archive for news.

◆ Count what survives

Many people meet their first fabricated citation inside the next five minutes.

This is the AI Lab's signature game: a stack of citations, some real, some fabricated, and your job is to catch every fake.

MODULE 5 OF 6

Deep research, and its blind spot

A 20-source report while you get coffee. Read this before you trust one.

What deep research agents actually do

- ◆ **You ask one question**

"Compare the evidence on X" or "What is the state of research on Y."

- ◆ **The agent browses for you**

It runs dozens of searches, opens dozens of pages, takes notes, and cross-references.

- ◆ **You get a cited report**

Pages long, structured, sourced, in minutes. Work that used to take a day.

This is real, and it is spectacular. ChatGPT, Gemini, Claude, and Perplexity all ship a version of it in 2026. Use them. Then verify like a pro.

Polish hides the holes

◆ The reports look finished

Clean sections, formal tone, numbered references. Everything signals "done."

◆ Reference checks say otherwise

2026 studies of deep-research agents keep finding unsupported claims and invented or misattributed sources inside otherwise-good reports.

◆ The failure hides in the middle

One weak source among nineteen real ones, wrapped in perfect formatting, is genuinely hard to spot.

The polish is the trap. A report that looks audited has not been audited. You are the audit.

A first draft with leads, not a finished answer

“Treat every deep-research report as a smart first draft assembled by a fast assistant who occasionally pads the bibliography.”

- ◆ **Use it for the map**

The report finds the terrain, the players, and the candidate sources faster than you ever could.

- ◆ **Do not outsource the conclusion**

The claims your decision rests on get verified by you, by hand, every time.

- ◆ **The workflow is: generate, then audit**

AI does the breadth. You do the depth, exactly where it matters.

People who skip the audit are not doing research with AI. They are doing vibes with citations.

Spot-check the load-bearing sources

◆ Find the conclusion's foundation

Which 2 or 3 sources does the report's main claim actually rest on? They are usually easy to spot.

◆ Run the 3-click check on those

Exist? Match? Say what the report says they say? Ten minutes, tops.

◆ Then decide how far to trust the rest

If the foundation holds, the report is a strong draft. If even one foundation source is fake or misread, distrust everything.

This single habit decides whether deep research tools make you faster and right, or just faster.

MODULE 6 OF 6

Stay honest, and when to trust

The last line of defense is not a better model. It is you.

Models improve. Judgment compounds.

◆ Every rate in this course will change

Fabrication rates fall, grounding improves, new tools ship monthly. Good.

◆ The habits do not expire

Receipts, lateral reading, opening citations, auditing the load-bearing sources. Those work on every model, forever.

◆ Judgment is the moat

Knowing what to check, and being willing to, is the research skill AI cannot replace. It is the one this certificate actually certifies.

The goal was never to distrust AI. It is to be the person whose work survives scrutiny because they checked.

Say when AI helped. Know the rules of the room.

- ◆ **There is no single rule, there are rooms**

A class syllabus, a journal's policy, a workplace norm. Each room decides what disclosure it expects.

- ◆ **School and research are strictest**

Many journals now require an AI-use statement. Many courses decide assignment by assignment. If unclear, ask first.

- ◆ **Everywhere, two things hold**

Never present an unverified AI claim as your own checked work, and never send what you have not read.

Disclosure costs one sentence. A fabricated citation with your name on it can cost a grade, a paper, or a reputation.

When to just trust the answer

◆ **Low stakes: trust freely**

Movie trivia, recipe swaps, brainstorming. Wrong is cheap. Enjoy the speed.

◆ **Medium stakes: check the one claim**

An email fact, a purchase decision. Sixty seconds on the load-bearing claim.

◆ **High stakes: full protocol**

Grades, money, health, law, reputation, anything published. Grounded tools, lateral reading, every citation opened.

Cost decides, not confidence. You already learned every move this table asks for.

The habits that compound

◆ Find well

Ask tools that show receipts. Retrieve-then-answer beats answer-from-memory.

◆ Verify well

Read laterally. SIFT. One independent source for every load-bearing claim.

◆ Cite well

Open every citation that matters. Three clicks. No exceptions for confident tone.

◆ Judge well

Deep research is a first draft. Disclose when AI helped. Let cost decide the depth of the check.

Screenshot this. Four lines, and they will outlive every model name in this course.

The researcher's pocket checklist

◆ Before you ask

Pick a tool with receipts for anything factual.

◆ When the answer arrives

Find the load-bearing claim. New tab. One independent source.

◆ Before you cite

Open it. Exists? Matches? Says it? Three clicks.

◆ Before you ship

Deep-research report? Audit the 2 or 3 foundation sources. Then disclose what AI did.

◆ Always

Confidence is style. Receipts are evidence. Cost decides how hard you check.

No tool names in this list, on purpose. Tools rotate. The checklist is permanent.

Where to go next, all free

- ◆ **SIFT and lateral reading, from the source**

Mike Caulfield's free materials teach the verification habit in depth.

- ◆ **Google Scholar**

Still the fastest way to confirm a paper exists and find who cited it.

- ◆ **Your AI's own deep research mode**

Run one real report, then audit it with this course's checklist. The practice is the lesson.

All verified this month. The full annotated list lives on the course resources page.

YOUR TURN

Find well. Verify well. Cite well.

◆ **Step 1, the lecture: done**

You just finished it by reading these slides.

◆ **Step 2: clear the 24 flashcards**

They are quick.

◆ **Step 3: the 3 AI Lab games**

Catch a fabricated citation with your own eyes. It changes you.

◆ **Then the quiz unlocks**

20 questions, 14 to pass, certificate prints on the spot.